

CS 202 Lab Assignment 11 (Total Marks= 20)

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Concurrent File Statistics Generator: To write a multithreaded Java program that reads multiple text files simultaneously using threads, counts the number of lines, words, and characters in each file, and writes the summary results into one common output file

[Marks =5 (Thread creation)+3 (File reading)+2x3 (Lines, words, char counting)+4 (Thread Sync)+2 (Appropriate Output)=20].

Problem Description: You need to accomplish the following tasks.

1. Take a list of text files (e.g., file1.txt, file2.txt, file3.txt).
2. For each file, create a separate thread that:
 - a. Reads the content using FileReader.
 - b. Counts:
 - i. Total number of lines
 - ii. Total number of words (split by spaces)
 - iii. Total number of characters.
3. Each thread writes its result into a shared file summary.txt.
4. Use synchronization to prevent file writing conflicts into the summary.txt file.
5. After all threads complete, print a message confirming the summary file is ready.